



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510 Fax: +66-2721-3515

SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510; Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Vitavax

Product name: carboxin 75% WP

Product use: Anilide fungicide

Relevant identified uses of the substance/mixture and uses advised against:

Used as a seed treatment for control of smut, rot, and blight on barley, oats, rice, cotton, vegetables, corn and wheat. It is also used to control fairy rings on turf grass. The product may be used to prevent the formation of these diseases or may be used to cure existing plant diseases.

2. HAZARD IDENTIFICATION

Manufacture, Storage and Import of Hazardous Chemicals Rules 1989

Classification: Not a hazardous substance or mixture

GHS Classification

Acute toxicity (Oral): Category 5

Acute toxicity (Inhalation): Category 4

Acute toxicity (Dermal): Category 5

GHS label elements

Hazard pictograms:



Signal word:

Warning

Hazard statements:

H303+H313 May be harmful if swallowed or in contact with skin.

H317 May cause an allergic skin reaction.

H332 Harmful if inhaled.

H373 May cause damage to organs through prolonged or repeated exposure if swallowed.

H410 Very toxic to aquatic life with long lasting effects.



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Precautionary statements:

Prevention:	P260 Do not breathe dust/fume/gas/mist/vapors/spray. P271 Use only outdoors or in a well-ventilated area. P272 Contaminated work clothing should not be allowed out of the workplace. P273 Avoid release to the environment. P280 Wear protective gloves.
Response:	P302+P352 IF SKIN: Wash with plenty of water. P304+P340+P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a Poison Center or doctor/physician if you feel unwell. P312 Call a Poison Center or doctor/physician if you feel unwell. P333+P313 If skin irritation or rash occurs: Get medical advice/attention. P362+P364 Take off contaminated clothing and wash it before reuse. P391 Collect spillage.
Disposal:	P501 Dispose of contents/container to an approved waste disposal plant.

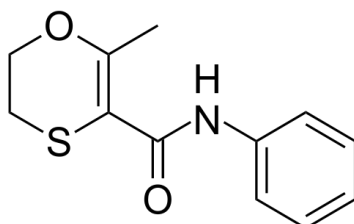
Other hazards which do not result in classification:

Occupational health effects due to inhalation of mineral dusts incorporating crystalline silica (quartz, cristobalite, tridymite), crystalline silicates (kaolin, talc) graphite or coal.

Avoid generating dust; fine dust dispersed in air in sufficient concentrations, and in the presence of an ignition source is a potential dust explosion hazard.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name:	2-methyl- <i>N</i> -phenyl-5,6-dihydro-1,4-oxathiine-3-carboxamide
Common name:	carboxin
Formula:	C ₁₂ H ₁₃ NO ₂ S
Molecular weight:	235.30 g/mol
Structural formula:	





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Ingredients and composition:

Component	CAS No.	Concentration (%W/W)
Active ingredient		
- Carboxin	5234-68-4	70 – 90
Other ingredient		
- kaolin	1332-58-7	1 – 10
- Distillates (petroleum), hydrotreated light naphthenic	64742-53-6	1 – 10
- Ethane-1,2-diol	107-21-1	1 – 10
- Sodium diisopropylnaphthalenesulphonate	1322-93-6	1 – 10
- Sodium 2-[methyleoleoylamino]ethane-1sulphonate	137-20-2	1 – 10

4. FIRST AID MEASURES

Description of first aid measures

General advice:

Move out dangerous area. Consult a physician. Show this safety data sheet to the doctor in attendance.

In case of swallowed:

Clean mouth with water and drink afterwards plenty of water. Do not give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Obtain medical attention.

In case of eyes contact:

Flush eyes with water as precaution. Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist.

In case of skin contact:

Take off contaminated clothing and shoes immediately. Wash off with soap and plenty of water.

In case of Inhaled contact:

Move to fresh air. Consult a physician after significant exposure.

Most important symptoms and effects, both acute and delayed:

Irritant effects. Sensitizing effects. Symptoms may be delayed. Harmful if inhaled. May cause damage to organ through prolonged or repeated exposure if swallowed.

Notes to physician:

The first aid procedure should be established in consultation with the doctor responsible for industrial medicine.



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5. FIRE FIGHTING MEASURES

Extinguishing Media:

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Specific hazards during firefighting:

Do not allow run-off from fire-fighting to enter drains or water courses.

Specific extinguishing methods:

Collect contaminated fire extinguishing water separately. This must not be discharged into drains.

Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

Special protective equipment for firefighters:

In the event of fire, wear self-contained breathing apparatus.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal precautions, protective equipment and emergency procedures:

Use personal protective equipment.

Avoid dust formation.

Avoid breathing dust.

Ensure adequate ventilation.

Environmental precautions:

Do not flush into surface water or sanitary sewer system.

Prevent further leakage or spillage if safe to do so.

If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up:

Pick up and arrange disposal without creating dust.

Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Advice on protection against fire and explosion:

Avoid dust formation. Provide appropriate exhaust ventilation at places where dust is formed.

Advice on safe handling:

Avoid formation of respirable particles.

Avoid exceeding the given occupational exposure limits (see section 8).

For personal protection, see section 8.



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Smoking, eating and drinking should be prohibited in the application area.

Dispose of rinse water in accordance with local and national regulations.

Conditions for safe storage:

Keep container tightly closed in a dry and well-ventilated place.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

Components with workplace control parameters:

Components	Value type (form of exposure)	Control parameters/ permissible concentration	basis
Kaolin	TWA (respirable fraction)	2 mg/m ³	ACGIH
Distillates (petroleum), hydrotreated light naphthenic	TWA (Mist)	5 mg/m ³	IN OEL
	STEL (Mist)	10 mg/m ³	IN OEL
	TWA (Inhalable fraction)	5 mg/m ³	ACGIH
Ethane-1,2-diol	C	100 mg/m ³	ACGIH
	C (Aerosol only)	100 mg/m ³	ACGIH

Personal Protective Equipment

Eyes Protection:

Eye wash bottle with pure water. Tightly fitting safety goggles.

Skin and Body Protection:

Dust impervious protective suit. Choose body protection according to the amount and concentration of the dangerous substance at the workplace.

Hand Protection:

Polyvinyl alcohol or nitrile- butyl-rubber gloves. The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.

Before removing gloves clean them with soap and water.

Respiratory Protection:

In the case of dust or aerosol formation use respirator with an approved filter.

Dust safety masks are recommended when the dust concentration is more than 10 mg/m³

Hygiene measures:

Handle in accordance with good industrial hygiene and safety practice.

When using do not eat or drink.

When using do not smoke.

Wash hands before breaks and at the end of workday.



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9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Solid
Color:	Blue
Odor:	Weak
pH:	Not applicable
Boiling point:	Not applicable
Evaporation rate:	No data available

10. STABILITY AND REACTIVITY

Reactivity:	Stable under recommended storage conditions.
Chemical stability:	No decomposition if stored and applied as directed.
Possibility of hazardous reactions:	Stable under recommended storage conditions. No decomposition if used as directed.
Conditions to avoid:	No data available
Incompatible materials:	None known
Hazardous decomposition products:	Carbon oxides Nitrogen oxides (NO _x) Sulphur oxides (SO _x)

11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, Carboxin:

LD50 (Oral, rat):	2,900 mg/kg
LD50 (Dermal, rabbit):	> 4,000 mg/kg
LC50 (Inhalation, rat, 4h):	> 4.7 mg/L
Skin irritation (rabbit):	Not irritant.
Eye irritation (rabbit):	Not irritant.
Skin sensitization:	May cause sensitization by skin contact in Guinea pig.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Adopt good working practices, so that the product is not released into the environment.



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Data referred to the ingredient, Carboxin:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h):	2.3 mg/L
Test type:	Flow-through test
GLP:	Yes

Daphnia magna (water flea),

EC50 (48 h):	> 57 mg/L
Test type:	Flow-through test
GLP:	Yes
NOEC (17 days):	0.32 mg/L
GLP:	Yes

Crab,

NOEC (21 days):	0.32 mg/L
GLP:	Yes

Algae – Acute toxicity:

Green algae,

EC50 (5 days):	0.48 mg/L
Test type:	Static test

Persistence and degradability

Data referred to the ingredient, Carboxin:

Biodegradability: Not readily biodegradable.

Bioaccumulative potential

Data referred to the ingredient, Carboxin:

Partition coefficient: $\log P_{ow} = 2.3$ (n-octanol/water)

Data referred to the ingredient, ethane-1,2-diol:

Partition coefficient: $\log P_{ow} = -1.3$ (n-octanol/water)

Mobility in soil

No data available

Other adverse effects

Data referred to the product:

Ozone-depletion potential: Regulation: 40 CFR Protection of Environment; Part 82 Protection of Stratospheric Ozone – CAA Section 602 Class I Substances.

Remarks: This product neither contains, nor was manufactured with a Class I or Class II ODS as defined by the U.S. Clean Air Act Section 602 (40 CFR 82, Subpt. A, App.A+B).



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Data referred to the ingredient, Carboxin:

Results of PBT and vPvB assessment: This substance is not considered to be persistent, bio-accumulating and toxic (PBT).

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues:

The product should not be allowed to enter drains, water courses or the soil.

Do not contaminate ponds, waterways or ditches with chemical or used container.

Offer surplus and non-recyclable solutions to a licensed disposal company.

Contaminated packaging:

Empty remaining contents. Dispose of as unused product. Do not re-use empty containers.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

International Regulations:

UNRTDG

UN No: 3077

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (carboxin)

Dangerous Goods Class: 9

Packaging Group: III

Labels: 9

IATA-DGR

UN No: 3077

Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (carboxin)

Dangerous Goods Class: 9

Packaging Group: III

Labels: Miscellaneous

Packing instruction (cargo aircraft): 956

Packing instruction (passenger aircraft): 956



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IMDG-Code

UN No: 3077
Proper shipping name: ENVIRONMENTALLY HAZARDOUS SEBSTANCE, SOLID,
N.O.S. (carboxin)
Dangerous Goods Class: 9
Packaging Group: III
Labels: 9
EmS Code: F-A, S-F
Marine pollutant: no

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture.

CAUTION

Harmful if inhaled. Avoid breathing vapor or spray mist.

Do not apply water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwaters or rinsate or by disposal of wastes.

This product may have effects on federally listed threatened or endangered species or their critical habitat in some counties. It is a violation of federal law to kill, harm or harass listed animal species without authorization. To limit the potential for such impacts when using this product, consult and follow the instructions provided in the EPA endangered Species bulletin for the Country or Parish in which you are applying the seed. To determine whether County or Parish has a bulletin, consult <http://www.epa.gov/espp> before each season's use of this product.

16. OTHER INFORMATION

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information releases only to the specific material designated and may not be valid for such material used in combination with any other materials in any process, unless specified in the text.